Same Responses, Different Questions

AP Statistics · Unit 2 · Topic 2.1 · TEACHER KEY

CED TETHER

- **Skill 2.D** — Compare distributions or relative positions of points within a distribution.
- **EU UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **LO UNC-1.P** — Compare numerical and graphical representations for two categorical variables. [Skill 2.D]
- **EK UNC-1.P.1** — Side-by-side bar graphs, segmented bar graphs, and mosaic plots are examples of bar graphs for one categorical variable, broken down by categories of another categorical variable.
- **EK UNC-1.P.2** — Graphical representations of two categorical variables can be used to compare distributions and/or determine if variables are associated.
- **EK UNC-1.P.3** — A two-way table, also called a contingency table, is used to summarize two categorical variables. The entries in the cells can be frequency counts or relative frequencies.
- **EK UNC-1.P.4** — A joint relative frequency is a cell frequency divided by the total for the entire table.

Essential question: Which display best answers a relationship question when the groups have different sizes?

DOK-3 skill: 4.B · **FRQ pattern:** `count-bars-vs-segmented-bars-which-answers-the-question`

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Chooses B for the relationship using 60 percent versus 30 percent
- **reasoning-2** (required) — Explains how equal whole-bar heights compare within-grade proportions
- **reasoning-3** (required) — Acknowledges Andre's group-size point and gives a count question A answers directly

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Comparing 72 with 18 as if those counts were within-grade percentages
- Saying equal-height bars mean the two grade distributions are the same
- Claiming Graph A contains no information about percentages rather than saying division is required
- Claiming Graph B shows 120 ninth graders and 60 twelfth graders

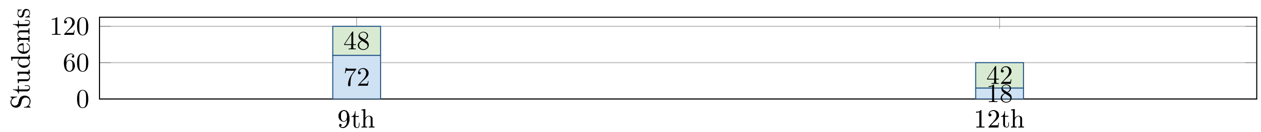
[STAR] THE PROBLEM — annotated

Suppose a student council asks 120 ninth graders and 60 twelfth graders whether they prefer Field Day or a Talent Show for the spring celebration. The same responses are shown in two graphs.

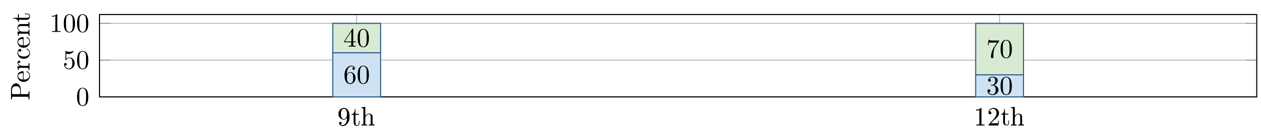
Andre: *Graph B hides how many people there are because both bars reach 100%.*

Leila: *Graph A hides the relationship because the grade totals are different.*

Graph A



Graph B



Lower: Field Day   Upper: Talent Show

FIRST TAKE — one sentence before you work the parts

For the question *Is celebration preference related to grade level?*, which graph would you use first? Name one visible feature.

Key: Not graded. The goal is to surface whether students instinctively compare bar totals, segment counts, or within-grade shares.

[BOOK] MATCH THE DISPLAY TO THE QUESTION (keep on the page)

A count bar graph shows numbers of students. A segmented bar graph stacks each group's percentages to 100%. To compare group preferences, divide each count by its group total and compare the within-group percentages. Different group distributions suggest association. Percentages alone do not show how many people were surveyed.

DOK 1 **(a)** From Graph A, state the total surveyed in each grade.

Key: There are 120 ninth graders and 60 twelfth graders.

DOK 2 **(b)** Find the percent who prefer Field Day in each grade. Show the fractions and compare the results.

Key: Ninth grade: $72/120 = 60\%$. Twelfth grade: $18/60 = 30\%$. Field Day preference is 30 percentage points higher in ninth grade.

DOK 3 **(c)** Which graph better shows whether grade and preference are related? Use both Field Day percentages and explain how the bar heights help. Then explain what Andre is right about and one question Graph A answers more directly. Write 3–4 sentences.

Key: Graph B shows the relationship more directly: its equally tall whole bars let us compare within-grade shares, 60% versus 30%, without unequal group sizes distorting the comparison. These different shares suggest an association. Andre is right that Graph B alone does not show how many students were surveyed. Graph A directly answers count questions such as how many twelfth graders chose Talent Show (42).